Arun Kumar MN

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Professional Summary

• Results-driven Technical Architect with 11+ years of experience in designing and building high-performance, scalable, and distributed applications using Java, Spring Boot, and Microservices architectures. Proven expertise in Reactive Programming (Spring WebFlux), GraphQL, and RESTful API design, with a strong focus on clean architecture, performance optimization, and system reliability.
• Extensive experience in database design and data modeling across SQL (Aurora PostgreSQL) and NoSQL (Cassandra, MongoDB) ecosystems, ensuring scalability and efficient data access patterns. Skilled in leveraging Kafka, Redis, and Elasticsearch for real-time data streaming, caching, and search optimization.
• Proficient in cloud-native application development on AWS, with hands-on experience in EC2, RDS, Lambda, Cognito, and IAM, and expertise in containerization (Docker, Kubernetes) and CI/CD pipelines for automated deployments. Adept at system design, design patterns, and object-oriented programming, ensuring maintainable and robust software solutions.
• Recognized for delivering high-quality enterprise solutions, fostering collaboration across teams, and driving innovation—awarded **Engineering Excellence and Product Excellence at Intuit** (2019–2020). A dedicated and detail-oriented professional committed to technical excellence and continuous learning.

Skills

Core Java, Spring boot 3.0, Spring webflux Reactive programming, GraphQl, Microservices, Restful APIs, Maven, Redis, Data Structures, Design Patterns, System design; SQL, Aurora postgres, NoSql MongoDB, Cassandra; Kafka, Elasticsearch, JUnit, TestNG, Integration testing, Jira, Swagger;AWS, Kubernetes, CI/CD; JDBC, Web development; Object Relational Mapping, Hibernate; HTML, CSS, JavaScript.
Tools: IntelliJ, Eclipse, TablePlus, SQL Developer, Kafka, Kibana, GIT, Jenkins, Splunk, Hubble, AWS, Tomcat, Jetty, Postman, Insomnia.

Work History

Technical Architect | Infosys (Client: Apple), 02/2024 to 10/2025
• Developed comprehensive solutions for Apple's software to manage customer engagement across international territories.
• Worked on Apple **disclosed projects** and **P0** applications enhancing customer and agent experiences.
• Designed and developed **reactive microservices** using **Spring Boot 3.0**, **Spring WebFlux**, and **Java 21** in a microservices architecture to build scalable and high-performance systems. Led technical design discussions and architectural reviews, influencing key implementation decisions and ensuring alignment with business goals.
• Designed new applications for **high transaction processing & scalability** in order to seamlessly support future modifications and growing volume of data processed in the environment.
• Modeled, designed, and optimized **Cassandra** database tables to support new features, ensuring efficient data partitioning, performance tuning, and schema evolution. Conducted proof-of-concepts (**POCs**) such as evaluating a secondary database strategy for Cassandra and assessing data replication and consistency trade-offs.
• Integrated **AI services** (Sierra and Chatkit) by developing an intermediate microservice to capture and publish bot response metadata for analysis, ensuring response authenticity and improving customer experience insights.
• Built and maintained **CI/CD pipelines** enabling automated build, testing, and deployment workflows for faster and reliable releases. Developed and executed JMeter **load** and **integration test** scripts to validate system performance under peak load and ensure production stability.
• Mentored junior and mid-level engineers through pair programming, code reviews, and guidance. Shared knowledge via tech talks, documentation, and design discussions.
• Deployed and managed applications across **Apple Cloud**, **AWS**, and **Apple Kubernetes (Kube)** clusters, ensuring zero-downtime deployments, auto-scaling, and service reliability. Managed non-production and production environments end-to-end, including host patching, infrastructure readiness, and application deployments.

Technical lead | Kalkitech (Full-Time), 08/2023 to 02/2024
• Led the design and development of end-to-end solutions for **Kalki IO,** implementing **IEEE 2030.5 (SEP 2.0)** standards for server and aggregator components.
• Architected and delivered microservice-based applications using **Spring Boot**, **Java**, **and RESTful APIs**, ensuring scalability and maintainability.
• Established coding standards, implemented CI/CD workflows, and integrated modern tools to streamline the development process.
• Collaborated with cross-functional teams and mentored developers, improving code quality, team efficiency, and delivery speed.

Senior Software Engineer | Intuit (Full-Time), 06/2019 to 08/2022
• Own & manage end to end software development to deliver high quality responsive and accessibility compliant web applications using Java + Spring Boot.
• Implemented **Reactive Programming** support for applications using **Spring webflux** and converted Spring application to Spring boot.
• Drove a **3x** increase in customer loan conversions by developing and launching new loan feature support, directly impacting business revenue.

• Added support on application to persist data in **NoSql db** and provided mechanism for pushing events to eventbus and then to data lake using internal system called Thrive mapping.
• Provided a new mechanism for onboarding company partners to Intuit systems using **MVEL** technology which reduced onboarding time from a few weeks to a week. Added **GraphQL** support for applications to handle schema based calls to the service which eliminated the need for storing client component structure in the database.
• Delivered a new line of business to Intuit through Quicken loans Refinance and Purchase experience which was a massive success in the loan department.
• **Received Engineering Excellence award and Product Excellence award for the year 2019-2020**.
• Delivered high quality insights to customers of Mint using Intuit's own Insight generation tool involving complex data queries run against millions of customer data, creating insights and delivering emails through notification engine.
• Handled **migration** and support for Quickbooks online system on **Aurora postgres** and designed infrastructure for running response comparison between old monolith and new **graphQL** endpoints of QBO.
• Implemented a robust system for converting non-Ansi SQL queries into Ansi postgres compatible queries.
• Developed innovative solutions to optimize database performance and reduce query times in order to improve overall system efficiency.
• Implemented continuous integration and continuous deployment (CI/CD) pipelines to automate the deployment process and ensure faster and more efficient delivery of software updates.

Lead Engineer / Programmer Analyst | Quest Global (Client: Apple), 07/2017 to 06/2019
• Led multiple engineering teams (4–6 developers) and took complete ownership of RSVP software modernization for Apple, driving end-to-end architecture, design, and delivery.
• Re-engineered existing services to improve performance by 15%, enhancing scalability and reliability across key AppleCare backend systems.
• Designed and implemented microservices-based architecture, ensuring modular, independent, and easily maintainable services with RESTful APIs.
• Integrated Kafka for asynchronous data processing and built a custom service bus to streamline communication between upstream and downstream systems.
• Created Spring applications testing suite using JUnit plugin and Mockito framework.
• Developed backend layers using Spring Boot, Spring MVC, JPA, and Hibernate, and implemented Spring Security, Shiro, and LDAP for authentication and authorization.
• Automated builds and deployments through Jenkins, Maven, and Ansible, enabling robust CI/CD pipelines and faster release cycles.
• Strengthened system observability by implementing Splunk logs, Hubble Metrics, and Orchestra for real-time monitoring and diagnostics.
• Collaborated with product owners, architects, and cross-functional teams to define requirements and deliver high-quality, scalable applications following Agile Scrum practices.
• Created comprehensive JUnit and Mockito test suites, ensuring quality and reducing regression defects across services.
• Played a key role in production release planning, issue resolution, and post-deployment performance optimization.
• Facilitated technical discussions, mentored developers, and guided decision-making to align solutions with architectural and business goals.

Software Developer | Quest Global, 07/2012 to 09/2014
• Contributed to all phases of the Software Development Life Cycle (SDLC) — including analysis, design, development, testing, and production support — using Java/J2EE and the Spring MVC framework.
• Designed and developed end-to-end web applications with Spring MVC, REST APIs, and Spring JPA (CRUD repositories) for data persistence and business logic implementation.
• Participated in data model design and migration activities, ensuring efficient data flow and schema optimization.
• Built user interfaces with HTML, CSS, JavaScript, and jQuery, incorporating client-side validation for improved usability and performance.
• Supported production systems, providing root-cause analysis and resolution for high-priority incidents, while coordinating with Tier 2 and Tier 3 support teams.
• Engaged in code reviews, shared best practices, and contributed to maintaining coding standards and design consistency across modules.
• Collaborated with UI/UX designers and backend teams to ensure cohesive, user-friendly, and scalable application design.
• Demonstrated ownership and accountability by ensuring timely delivery of high-quality, maintainable solutions aligned with project requirements.